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Inventor(s)	Dominguez; Alan et al.

Long-term medicament storage apparatus and methods

Abstract

An apparatus for storing a medicament delivery device, the apparatus comprising: a container defining a storage cavity having an opening, the storage cavity configured to receive a syringe pre-filled with medicament; an absorbent disposed in the cavity, the absorbent at least partially hydrated with a liquid solution; and a seal member at least selectively connected with the cavity to form a seal that is at least substantially gas impermeable.

Inventors: **Dominguez; Alan (Thousand Oaks, CA), Clark; Nicholas J. (Thousand Oaks, CA)**

Applicant: **AMGEN INC.** (Thousand Oaks, CA)

Family ID: **1000008781935**

Assignee: **AMGEN INC. (Thousand Oaks, CA)**

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Primary Examiner: Ackun; Jacob K

Attorney, Agent or Firm: MARSHALL, GERSTEIN & BORUN LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is the United States national phase of International Patent Application No. PCT/US2021/048984, filed Sep. 3, 2021, which claims priority to United States Patent Application No. U.S. 63/090,500, filed Oct. 12, 2020, the entire contents of each of which are hereby expressly incorporated herein by reference.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates to medicament storage apparatus and, more particularly, to long-term, humidity-controlled medicament storage apparatus and related methods.
BACKGROUND
(2) Pre-filled medicament treatments can be used for home-use because the treatments can be prepared with a required, single-use dosage of a pharmaceutical product. Some pre-filled medicament treatments can include pre-filled hypodermic syringes or autoinjector products, for example. These pre-filled medicament treatments are made to be easily used by patients.
(3) Such pre-filled medicament products can often be stored for extended periods of time, sometimes longer than two years. Although the pre-filled medicament products include various shields and seals both at the rear of the primary container and at the open needle tip, in certain environments, the medicament could potentially dry out and clog the needles of the pre-filled medicament products. When the needle clogs, the pre-filled medicament product may be more difficult to use.
SUMMARY
(4) Disclosed herein is an apparatus for storing a medicament delivery device long term. The apparatus includes a container defining a storage cavity having an opening, the storage cavity configured to receive a syringe pre-filled with medicament. Additionally, the container includes an absorbent disposed in the cavity, and the absorbent is at least partially hydrated with a liquid solution. The apparatus further includes a seal member at least selectively connected with the cavity to form a seal that is at least substantially gas impermeable.
(5) Additionally disclosed herein is a method of packaging a medicament delivery device for storage. The method includes providing a container having a cavity and an opening, and disposing a syringe pre-filled with a medicament in the cavity of the container. Additionally, the method includes disposing an absorbent in the cavity of the container, the absorbent being at least partially hydrated. Further, the method describes sealing the cavity of the container with a seal member by at least selectively connecting the seal member to the cavity to form a seal that is at least substantially gas impermeable.
(6) Additionally disclosed herein is an apparatus for storing a medicament delivery device. The

apparatus includes a rigid storage container configured to receive at least one prefilled medicament delivery device or at least one container including a prefilled medicament delivery device. The apparatus also includes an absorbent carrier disposed in the rigid storage container, the absorbent carrier including an absorbent at least partially hydrated with a liquid solution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) It is believed that the disclosure will be more fully understood from the following description taken in conjunction with the accompanying drawings. Some of the drawings may have been simplified by the omission of selected elements for the purpose of more clearly showing other elements. Such omissions of elements in some drawings are not necessarily indicative of the presence or absence of particular elements in any of the exemplary embodiments, except as may be explicitly delineated in the corresponding written description. Also, none of the drawings are necessarily to scale.

(2) FIG. **1** is a side view of a pre-filled hypodermic syringe including a needle shield.

(3) FIG. **2** is a perspective view of a container for storing a medicament delivery device in accordance with the present disclosure, such as the pre-filled hypodermic needle of FIG. **1**.

(4) FIG. **3** is a perspective view of the container for storing a medicament delivery device of FIG. **2**, further including a gas permeable seal member disposed over a first sub-cavity.

(5) FIG. **4** is a perspective view of the container for storing a medicament delivery device of FIG. **3**, further including a member disposed over the container.

(6) FIG. **5** is a perspective view of an alternative apparatus for storing a medicament delivery device in accordance with the present disclosure.

(7) FIG. **6** is an example graph of relative humidity in an enclosed container based on a hydration time for an absorbent.

(8) FIG. **7** is an example graph of relative humidity in an enclosed container based on hydration of an absorbent with a water and alcohol solution.

(9) FIG. **8***a* is a perspective view of a rigid storage container in accordance with the present disclosure.

(10) FIG. **8***b* is an additional view of a graphical display disposed on the rigid storage container of FIG. **8***a*.

(11) FIG. **8***c* is an additional view of a graphical display disposed on the rigid storage container of FIG. **8***a*.

(12) FIG. **8***d* is a fingerprint lock interface disposed on the rigid storage container of FIG. **8***a*.

(13) FIG. **9** is a perspective view of the rigid storage container of FIG. **8***a* in an open state for storing at least one container for storing a medicament delivery device.

(14) FIG. **10** is a perspective view of a container for storing one container having a medicament delivery device in accordance with the present disclosure.

(15) FIG. **11***a* is a perspective view of a rigid, travel container in an open state for storing at least one medicament delivery device in accordance with the present disclosure.

(16) FIG. **11***b* is a perspective view of a rigid, travel container in a closed state for storing at least one medicament delivery device.

(17) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments. It will

further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

DETAILED DESCRIPTION

(18) Pre-filled medicament delivery devices are provided to patients for their personal administration at home, for example. Even when stored under ideal conditions, the pre-filled medicament delivery device can dry out and become clogged. When the medicament in the pre-filled medicament delivery device is dried or clogged, it may be difficult or painful to administer the medicament in the pre-filled medicament delivery device.

(19) The apparatus for storing medicament delivery devices of the present disclosure inhibits the drying out of medicament during long-term storage. For example, the apparatus may include an absorbent designed to maintain an elevated relative humidity in an enclosed container. The absorbent is any material that can be hydrated, store moisture, and release the moisture to maintain a humid environment in an enclosed area. Additionally, the absorbent can be optimally designed based on the size of the container, a desired relative humidity level, and the length of storage, etc.

(20) Additionally, the apparatus and method of using the apparatus can include sterilizing the apparatus for storing medicament delivery devices, including a container, the medicament delivery device, and the absorbent. For example, the apparatus can be sterilized using a sterilant such as a vapor pressure hydrogen peroxide (VPHP, also referred to as vaporized hydrogen peroxide (VHP)) process and/or hydrate the absorbent in a water sterilants solution, such as an ethanol, isopropanol, or benzyl alcohol solution. As a result, the apparatus for storing the medicament delivery device can be sterile and safe for personal use when the patient opens the apparatus for storing medicament delivery devices to self-administer a medicament.

(21) FIG. **1** is a side view of a pre-filled hypodermic syringe **100** including a needle shield **102**. The pre-filled hypodermic syringe **100** includes a container **110** in fluid communication with a needle **112**. The needle shield **102** covers the needle **112** and protects the needle **112** from an exterior environment. The sealed container **110** is filled with a dosage of medicament **116**. The container **110** is filled with a predetermined amount of medicament **116**, that, for example, is suitable for convenient personal use by patients. As shown in FIG. **1**, the pre-filled hypodermic syringe **100** and the needle shield **102** are transparent, however, in other examples, the pre-filled hypodermic syringe **100** and/or the needle shield **102** is opaque.

(22) The needle **112** is in fluid communication with the container **110** and an exterior environment. The needle **112** is disposed within the needle shield **102**. The needle shield **102** can be composed of plastic, rubber, silicone or similar material capable of protecting the needle **112** during transportation and storage and covering the needle **112** from the surrounding environment to ensure the needle **112** is sterile. In some examples, the needle shield **102** can be nonporous material to completely seal the needle **112** from the surrounding environment or the needle shield **102** can be a porous material that partially seals the needle **112** from the surrounding environment, because the needle shield **102** is gas permeable.

(23) The medicament **116** may be subject to drying out and can be more difficult to use. For example, the needle shield **102** is porous enough to allow moisture from the medicament to evaporate. As a result, medicament **116** that may reside in the needle **112** can at least partly solidify, and, can partially or entirely clog the needle. When the needle is partially clogged in a hypodermic syringe, the syringe plunger may be difficult and painful to actuate. For example, a syringe plunger is difficult to use if actuating the plunger requires approximately 40 Newtons (N) of force or more.

(24) Similarly, medicament **116** can be stored and dry out in an autoinjector product, such as an autoinjector cartridge. As a result, the medicament **116** can still dry out and clog a needle carried by

the autoinjector cartridge. Accordingly, an autoinjector needle can become partially or entirely clogged. When the needle is partially or entirely clogged, the autoinjector may, when initiating a medicament delivery, provide an error message that the force required to deliver the medicament **116** exceeded a preset threshold. For example, autoinjectors are designed to delivery between 20 and 120 Newtons (N) of force.

(25) FIG. **2** illustrates an embodiment of the medicament storage apparatus **200**. FIG. **2** provides a perspective view of the apparatus **200** including a container **202** for storing a medicament delivery device **204**, such as the pre-filled hypodermic needle **100** of FIG. **1**. As shown in FIG. **2**, the apparatus additionally includes an absorbent **206** in the container **202**. The container **202** provides long-term protection and storage for the medicament delivery device **204**.

(26) The container **202** includes a container edge **208** circumscribing an opening **210** and a storage cavity **212**. The storage cavity **212** is configured to receive the medicament delivery device **204**, such as a syringe **214** pre-filled with medicament. The syringe **214** additionally includes a needle shield **216** and a syringe plunger **218**, similar to the pre-filled hypodermic syringe **100** of FIG. **1**. As shown in FIG. **2**, the container **202** is transparent, however the container **202** can be translucent, cloudy, or opaque. For long-term storage, the container **202** is preferably made of a rigid, gas impermeable material, such as plastic, to seal the environment within the container **202** and protect the syringe **214** from damage during transportation and storage.

(27) The absorbent **206** is also disposed in the storage cavity **212** of the container **202**. The absorbent **206** is at least partially hydrated with a liquid solution, such as a water solution. Alternatively, the liquid solution can be a water and alcohol solution to maintain the sterility of the container **202** during long-term storage. The absorbent **206** can be a humidification pouch, a humidor bag, a sponge, a fabric article, liquid gels, or similar liquid absorbent material that also allows the liquid to evaporate over time. Depending on the desired humidity in the container **202** and the absorbent **206** material, the absorbent may be soaked in the liquid solution for a predetermined period of time (e.g., 1 second, 5 seconds, 10 seconds, 20 seconds, etc.). Alternatively, the absorbent may be provided a specified volume of liquid solution (e.g., 0.5 milliliter (mL), 1 mL, 2 mL, 5 mL) or a specified weight of liquid solution (e.g., 0.5 grams (g), 1 g, 2 g, 5 g). Furthermore, in some examples, the absorbent may be provided liquid solution proportional to the dry weight of the absorbent (e.g., 50 percent (%) dry weight of liquid solution, 100%, 150%). Accordingly, any method of measuring an amount of liquid absorbed by the absorbent could be considered within the scope of the disclosure.

(28) The storage cavity **212** and its contents are sterile, including the absorbent **206** and syringe **214**. Before and/or after placing the syringe **214** and the absorbent **206** in the container **202**, the storage cavity **212** may additionally undergo a sterilization process. Sterilizing the storage cavity **212** and the syringe **214** may include exposing the storage cavity **212** and the syringe **214** to a sterilant. For example, the container and its contents can undergo a vapor phase hydrogen peroxide (VPHP) 5

(29) sterilizing process, however other medical storage sterilization methods are also considered within the scope of the disclosure.

(30) The storage cavity **212** of the container **202** is divided into a first sub-cavity **220** and a second sub-cavity **222** separated by a container wall **224**, the first sub-cavity **220** containing the syringe **214** and the second sub-cavity **222** containing the absorbent **206**. In other examples, the storage cavity **220** could include additional container walls to divide the storage cavity **220** into more than two sub-cavities. While the absorbent **206** is shown in the second sub-cavity **222**, the absorbent can be disposed in the first sub-cavity **220** along with the syringe **214**. The first sub-cavity includes a first retainer **226** and a second retainer **228** that receive the syringe **214**. The first retainer **226** and the second retainer **228** can additionally secure the syringe **214** and inhibit or reduce movement of the syringe **214** within the first sub-cavity **220**.

(31) FIG. **3** is a perspective view of the container **202** for storing a medicament delivery device **204**

of FIG. **2**, further including a gas permeable seal member **302** disposed over the first sub-cavity **220**. Gases and vapors can pass through the gas permeable seal member **302**. But the gas permeable seal member **302** prevents larger particles and materials from passing into or out of the first sub-cavity **220**, including bacteria and fungus spores. Gas permeable seal member **302** can include any of a variety of gas permeable seal members well known in the field of sterile medical packaging.

(32) As shown in FIG. **3**, the gas permeable seal member **302** is disposed on the container **202** over at least part of the opening **210** enclosing at least part of the first sub-cavity **220** containing the syringe **214**. The gas permeable seal member **302** allows gas and vapor to pass through, but protects the syringe **214** from bacteria, mold, and other particulates. Thus, the gas permeable seal member **302** maintains the sterility of the syringe **214** for long-term storage. In some embodiments, the gas-permeable seal member **302** can additionally cover the second sub-cavity **222**.

(33) As shown in FIG. **3**, the absorbent **206** is disposed in the second sub-cavity **222**, outside the gas permeable seal member **302**. Even though the first sub-cavity **220** is covered by the gas permeable seal member **302**, moisture that evaporates from the absorbent **206** can fill the entire container. Because the gas permeable seal member is permeable to water vapor, moisture from the absorbent **206** fills the entire cavity, including the first sub-cavity **220**.

(34) FIG. **4** is a perspective view of the container **202** for storing a medicament delivery device **204** of FIGS. **2** and **3**, further including packaging **402** and a seal member **404**, which in some versions can be a gas impermeable seal member, disposed over the container **202**. The packaging **402** including the container **202** is configured for long-term storage, sometimes in excess of two years. Additionally, the packaging **402** can include instructions for use **410** or other information useful to the end user or patient.

(35) The seal member **404** is disposed adjacent the container **202** and over the opening **210** to enclose the entire storage cavity **212**. Accordingly, the gas permeable seal member **302** is between the seal member **404** and the container **202**. The seal member **404** forms a gas impermeable seal between the seal member **404** and the container **202**. As a result, the container **202** is an enclosed environment, and the sterility of the container **202** is maintained as long as the seal is maintained.

(36) The absorbent **206** is hydrated and disposed in the second sub-cavity **222** before the seal member **404** is disposed on the container. The absorbent **206** releases moisture to maintain the humidity within the sealed container **202** at a desired level. As a result, if the container is maintained at a high relative humidity, medicine in the medicament delivery device **204** will not dry out nor get clogged. The absorbent **206**, disposed in the second sub-cavity, maintains a high relative humidity (e.g., greater than 60% relative humidity) throughout the entire container **202**. For example, the relative humidity can be maintained above 75% relative humidity for extended periods of time (e.g., 1 month, 6 months, 1 year, 3 years). Thus, the absorbent **206**, releasing maintaining humidity in the container, can inhibit clogging of the medicament delivery device **204** for months or years.

(37) FIG. **5** illustrates an additional embodiment of a medicament storage apparatus. FIG. **5** provides a perspective view of an alternative apparatus **500** for storing the medicament delivery device **204**. In contrast to the apparatus **200** of FIGS. **2**, **3**, and **4**, the apparatus **500** of FIG. **5** includes a container **510** disposed in a sealed gas-impermeable bag **512**. The gas-impermeable seal member is a sealed gas-impermeable bag **512** in which the container **510**, the syringe **514**, and the absorbent **516** are disposed.

(38) The container **510** is constructed from a rigid material, capable of protecting the medicament delivery device **204**, such as a syringe **514**, during storage and transportation. Additionally, the container **510** includes a first retainer **520** and a second retainer **522**. The first retainer **520** and the second retainer **522** receive, and in some examples, secure the syringe **514** from movement within the container **510**. Additionally, the container **510** can include a seal member **534** that can be either gas permeable or gas impermeable. As a result, the absorbent **516** may maintain the humidity of the entire gas-impermeable bag **512** or just the container **510**.

(39) Additionally, the container is placed within the gas-impermeable bag **512**. The gas-impermeable bag **512** is sealed to maintain a sterile environment including the container **510**. The gas-impermeable bag **512** may undergo a sterilization procedure (e.g., the VPHP process) while the container **510** is disposed in the gas-impermeable bag **512**. After being sealed, the gas-impermeable bag **512** needs to be cut or torn open to remove the container **510**. In some preferred embodiments, the gas-impermeable bag **512** is designed to be cut for arthritic patients that may not prefer tearing the gas-impermeable bag **512** open.
(40) As shown in FIG. **5**, the absorbent **516** is disposed near a needle shield **530** of the syringe **516**, but can be disposed anywhere within the container **510**. The absorbent **516**, disposed in the container **510**, maintains a high relative humidity (e.g., greater than 60% relative humidity) throughout the entire container **510**. Accordingly, the absorbent **516**, disposed in the container **510**, can inhibit the clogging of a medicament delivery device, such as the syringe **514**.
(41) FIG. **6** is an example graph **600** of relative humidity in an enclosed container based on a hydration time for an absorbent. The horizontal axis relates to a time in hours from zero (0) hours to 625 hours (approximately 26 days). Additionally, the vertical axis relates to measurements of relative humidity in the enclosed container from 40% to 100% relative humidity. The graph **600** incudes a first dataset **610**, a second dataset **620**, and a third dataset **630** that measure the relative humidity in the enclosed container over time.
(42) As shown in FIG. **6**, the first dataset **610** corresponds to the relative humidity for an absorbent hydrated for 5 seconds. The first dataset **610** shows humidity fluctuating between 70% and 80%. In contrast, the second dataset **620** (corresponding to 10 seconds of hydration) and the third dataset **630** (corresponding to 15 seconds of hydration) both keep the container at a relative humidity between 95% and 100%. Accordingly, hydration time can be used to control the relative humidity in the container over extended periods of time.
(43) Although each dataset of graph **600** is measured in hydration time, the datasets could be measured in weight, volume, or other measurement of stored liquid. For example, for some absorbents, 5 seconds of hydration (corresponding to the first dataset **610**) could equal approximately 6 grams (g) of water or approximately 6 milliliters (mL) of water while 15 seconds of hydration (corresponding to the third dataset **630**) could equal approximately 9 g of water or 9 mL of water. Additionally, hydration for 10 minutes could only absorb 16 g (or 16 mL) of water. Other absorbents may absorb a weight or volume of water faster or slower.
(44) The absorbent can be hydrated for a preset period of time based on design considerations, including the size of the container, the desired relative humidity, and the permeability of the needle shield covering the needle. Typically, the absorbent is hydrated for a period of time between approximately 5 seconds and approximately 10 seconds to maintain a relative humidity between 75% and 95%. As indicated by the first dataset **610**, the absorbent maintained a relative humidity between 70% and 80% for the duration of the experiment. Accordingly, longer hydration time would result in a higher relative humidity maintained in the container.
(45) Absorbents of different designs may require more or less hydration time to maintain a similar relative humidity. Additionally, absorbents placed in larger containers may require more hydration than similar absorbents placed in smaller containers. Furthermore, if the medicament delivery device includes a nonporous needle shield, the relative humidity does not need to be as high if the needle shield were more porous, because the nonporous needle shield inhibits the drying of the medicament. Each of the above factors can be considered when designing the apparatus of the present disclosure.
(46) FIG. **7** is an example graph **700** of relative humidity in an enclosed container based on hydration of an absorbent with a water and alcohol solution (i.e., ethanol alcohol). The horizontal axis relates to a time in hours from zero (0) hours to 520 hours (approximately 21 days) of measuring the relative humidity in the container. Additionally, the vertical axis relates to measurements of relative humidity in the enclosed container from 40% to 100% relative humidity.

The absorbent can by hydrated with other liquid solutions other than water or water and ethanol alcohol, such as benzyl alcohol or other sterilants.

(47) As shown in FIG. **7**, an absorbent is hydrated in an ethanol water solution (EtOH) for 15 seconds. Ethanol is often used in antibacterial applications, and could be added to the hydration solution to further maintain the sterile environment of the container during long-term storage. Similar chemicals could be added to the hydration solution to maintain a desired relative humidity and a sterile environment in the container, such as isopropanol or benzyl alcohol solutions. The percentage of ethanol in the ethanol-water solution appears to provide minimal effect on the relative humidity maintained in the container.

(48) As shown in FIG. **7**, the first dataset **710** corresponds to an absorbent soaked in a 20% EtOH solution for 15 seconds. The second dataset **720** corresponds to an absorbent soaked in a 50% EtOH solution for 15 seconds. And the third dataset **730** corresponds to an absorbent soaked in a 70% EtOH solution for 15 seconds. As shown in FIG. **7**, each of the first dataset **710**, the second dataset **720**, and the third dataset **730** show the humidity in the container is maintained above 95%. Thus, the graph **700** indicates the percent of ethanol in the EtOH solution has little effect on the humidity maintained in the container.

(49) Although each dataset of graph **700** is measured in hydration time, the datasets could be measured in weight, volume, or other measurement of stored liquid. For example, for some absorbents, 15 seconds of hydration could equal approximately 1.5 grams (g) of EtOH or approximately 1.6 milliliters (mL) of EtOH solution. Similarly, hydration for 10 minutes might only equal approximately 2 g of EtOH or 2.2 mL of EtOH. Other absorbents may absorb a weight or volume of water faster or slower.

(50) FIG. **8***a* is a perspective view of a rigid storage container **800** made in accordance with the present disclosure. The rigid storage container **800** is a container made of a rigid material and includes a storage cavity. The rigid container **800** includes a bottom **802** and a top **804**. Both the bottom **802** and the top **804** fully enclose the rigid storage container **800**. The rigid storage container **800** can be made of a rigid plastic, metal, or similar material that can support the weight of the rigid storage container **800** and protect the contents of the rigid storage container **800**. In some examples, the rigid storage container **800** can additionally include a handle for easier carrying of the rigid storage container **800**.

(51) The rigid storage container **800** is preferably designed to be a sleek and slim space saving design, so as to minimize the amount of space the rigid storage container uses in a patients refrigerator. Additionally, the outer casing of the rigid storage container **800** is made of a durable plastic or a stainless steel that can have antimicrobial properties. Other materials are possible. The casing may also be of a double-walled construction to improve the insulation properties of the rigid storage container **800**. However, in the event of power failure or extended storage outside of a refrigerator, the rigid storage container **800** may include a coolant or alcohol filled compartment (not shown). The coolant or alcohol compartment passively cools the rigid storage container **800** via the coolant or alcohol filled compartment. Thus, the rigid storage container **800** can, for short durations, be cooled if the temperature within the rigid storage container is rising above preset thresholds. Alternatively, the rigid storage container **800** includes battery powered apparatus, including the coolant or alcohol filled compartment, to allow for battery powered cooling of the rigid storage container **800**.

(52) The rigid storage container **800** includes a locking mechanism **806** and a graphical display **808**. Both the locking mechanism **806** and the graphical display **808** are powered by batteries (not shown) stored within the rigid storage container **800**. In other examples, the rigid storage container **800** could be powered via an alternative electrical sourced, including being plugged into an electrical outlet or having a photovoltaic cell in addition to batteries. As shown in FIG. **8***a*, the locking mechanism **806** and the graphical display **808** are disposed on a sidewall of the bottom **802**. In other examples, the locking mechanism **806** and/or the graphical display **808** are disposed

on the top **804** or elsewhere on the bottom **802**. The graphical display **808**, as shown, displays a temperature value corresponding to the temperature of the interior of the rigid storage container **800**. Alternatively, the graphical display **808** can display a humidity corresponding to the relative humidity of the interior of the rigid storage container **800**. As such, the container **800** can include a temperature sensor and/or a humidity sensor (not shown) disposed inside of the container and communicatively coupled to the graphical display **808**. Other sensors can also be included.

(53) FIG. **8***b* is an additional view of the graphical display **808** disposed on the rigid storage container **800** of FIG. **8***a*. The graphical display **808** includes a first display **810**. As shown in FIG. **8***b*, the first display **810** additionally includes a graphic indicating a temperature within the rigid storage container **800** of FIG. **8***a*. The example first display **810** shows the internal temperature of the rigid storage container **800** is now 44 degrees Fahrenheit (° F.). The graphical display **808** additionally includes a color indicator **812**. The color indicator **812** corresponds to the information provided on the first display **810**. For example, the color indicator **812** can display a color indicating a positive condition, such as green to indicate the temperature is within an appropriate or target range of temperatures. Additionally, the graphical display **808** includes a display toggle **814**. The display toggle **814** can be pressed to activate the graphical display **808** or alternate the graphical display between temperature, humidity, remaining battery life, or hours until temperature excursion. Drug and medicament are designed to be stored within a set range of storage temperatures. Temperature excursion is when the medicament or drug in the drug delivery device is exposed to temperatures outside the range of storage temperatures for the drug or medicament. In some examples, the graphical display **808** can include more or fewer display toggles **814** than shown in FIG. **8***b*.

(54) FIG. **8***c* is an additional view of the graphical display **808** disposed on the rigid storage container **800** of FIG. **8***a*. The graphical display **808** of FIG. **8***c* includes a second display **812**, different from the first display **810** of FIG. **8***b*. As shown in FIG. **8***c*, the interior temperature of the rigid storage container **800** is now 47 degrees Fahrenheit (° F.) and the color indicator **812** has changed to a second color. For example, the color indicator **812** can transition from green associated with appropriate or target range (i.e., 44° F.) to red associated with the higher 47° F., which is outside of the appropriate or target range. In other examples, the color indicator **812** can change based on an internal relative humidity, an amount of remaining battery life, or hours until temperature excursion.

(55) FIG. **8***d* is a fingerprint lock interface **806** disposed on the rigid storage container **800** of FIG. **8***a*. The fingerprint lock interface **806** inhibits tampering with the contents of the rigid storage container **800**. The fingerprint lock interface **806** recognizes a fingerprint of a patient or other authorized user and unlocks the rigid storage container **800**. The fingerprint lock interface **806** includes a fingerprint reader **820** and a first toggle **822** and a second toggle **824**. The fingerprint reader **820** scans a fingerprint placed on the fingerprint reader **820** and verifies if the fingerprint matches the fingerprint of the patient or other authorized user. Additionally, if needed, the first toggle **822** and the second toggle **824** can be used to control the fingerprint lock interface **806**.

(56) In other embodiments the fingerprint lock interface **806** could be a number pad for entering a numerical pin or a key hole for use with a key, or any similar locking mechanism. Additionally, the rigid storage container **800** can include two or more locking mechanisms. For example, a mechanical locking mechanism could be used if the batteries in the rigid storage container **800** cannot effectively power the fingerprint lock interface **806**.

(57) FIG. **9** is a perspective view of the rigid storage container **800** of FIG. **8***a* in an open state. As shown in FIG. **9**, the rigid storage container **800** opens along hinges **902** to show the storage cavity of the rigid storage container **800**. The top **804** of the rigid storage container **800** additionally includes a latch **904**. The latch **904** engages with the fingerprint lock interface **806** to secure the top **804** in a closed position. When the top **804** is in the closed position and the latch **904** is engaged with the fingerprint lock interface **806**, the rigid storage container **800** forms a sealed interior

container.
(58) The rigid storage container **800** additionally includes an absorbent carrier **910**. The absorbent carrier **910** includes a base **912** and a gas permeable lid **914**. The gas permeable lid **914** allows moisture stored in an absorbent **916** to pass therethrough. The absorbent **916** is similar to the absorbent **206**. For example, the absorbent **916** can absorb similar water solutions as absorbent **206** and can be made of similar materials as absorbent **206**. The gas permeable lid **914** can be opened so the absorbent **916** can be removed and rehydrated or replaced, for example, by the patient or other end user. In some examples, the absorbent **916** is antimicrobial to impede microbe growth during storage. The rigid storage container **800** further includes edge **918**, for enclosing the inner container of the rigid storage container **800** from the exterior environment. In some examples, the edge **918** closes against an inner surface of the top **804** or an edge **919** of the top **804**. Additionally the edge **918** and/or the edge **919** may include an elastic material to seal the container **800** from the exterior environment.
(59) The rigid storage container **800** is designed for storing at least one container **920** for storing a medicament delivery device. As shown in FIG. **9**, the rigid storage container **800** includes a first container **920***a*, a second container **920***b*, a third container **920***c*, and a fourth container **920***d*. In other examples, the rigid storage container **800** includes more or fewer containers than four. Each of the containers **920***a*, **920***b*, **920***c*, and **920***d* are substantially identical to the container **920**. Container **920** is a blister pack having a tray **922** and a gas permeable film **924** enclosing the container **920**. Contained within the container **920** is a prefilled medicament delivery device **926**. In some examples, to access the prefilled medicament delivery device **926**, the film **924** is peeled away from the tray **922**. Alternatively, the prefilled medicament delivery device **926** can be pushed through the film **924** through the tray **922**. In some examples, the prefilled medicament delivery device **926** is a prefilled syringe and the container additionally includes a syringe plunger **928**. Each of the containers **920***a*, **920***b*, **920***c*, **920***d* container substantially identical contents.
(60) FIG. **10** is a perspective view of prescription packaging **1000** including a container **1002** for storing one container **1004** and an absorbent **1006**. The container **1002** of the prescription packaging **1000** is sealable with a seal **1012**. In some examples, the container **1002** is a gas impermeable bag and the seal **1012** is resealable. The container **1002** additionally includes a label **1014** and instructions for use **1016** (IFU). The container **1002** can include the absorbent **1006**, similar to the absorbent **206** and the absorbent **916**. As a result, when the container **1002** is sealed, the absorbent **1006** maintains a desired relative humidity within the container **1002**.
(61) The prescription packaging additionally includes the container **1004** and is substantially similar to the container **920** of FIG. **9**. The container **1004** includes a tray **1022** and a film **1024** to seal the container **1004**. The container **1004** additionally includes the prefilled medicament delivery device **1026**. In some examples, the prefilled medicament delivery device **1026** is a prefilled syringe and the container **1004** additionally includes a plunger **1028** to actuate the prefilled syringe. In some examples, the film **1024** is a gas permeable film that allows moisture to pass through the film **1024**. In other examples, the film **1024** is not gas permeable and moisture cannot pass through the film **1024**. In such examples, the absorbent **1006** is disposed within the container **1004**.
(62) FIG. **11***a* is a perspective view of a rigid, travel container **1100** in an open state for storing at least one medicament delivery device. The travel container **1100** includes a bottom container **1102** and a lid **1104**. The lid **1104** includes a latch **1106** to close the travel container **1100**. The travel container **1100** is designed to store an absorbent carrier **1108** and at least prefilled medicament delivery device. As shown in FIG. **11***a*, the travel container **1100** includes a washable foam insert **1110** configured to receive a first prefilled medicament delivery device **1112***a* and a second prefilled medicament delivery device **1112***b*. In other examples, the travel container **1100** is designed to container more or fewer prefilled medicament delivery devices.
(63) The travel container **1100** is designed to store and transport medicament delivery devices for short or long term travel. Accordingly, the travel container **1100** is designed to maintain a humid

environment within the travel container for extended periods of time (e.g., days, weeks, months). As a result, the travel container creates a sealed environment and the absorbent carrier **1108** includes an absorbent to maintain humidity within the travel container when sealed. The absorbent carrier **1108** is similar to the absorbent carrier **910** of FIG. **9**. For example, the absorbent carrier **1108** can be opened and rehydrated by a patient or other end user. Additionally, the absorbent carrier **1108** preferably includes antimicrobial properties to maintain sterility of the travel container **1100**.

(64) FIG. **11***b* is a perspective view of the rigid, travel container **1100** in a closed state for storing at least one medicament delivery device. The travel container **1100** includes a prescription label **1120**. The prescription label **1120** is substantially similar to prescription label **1014**. As shown in FIG. **11***b*, the latch **1106** can be used to both seal the container **1100** and also, when pulled, can open the container **1100**. The rigid, travel container **1100** is designed to be small and sleek, for convenience during packing and short term travel storage.

(65) While the apparatus and methods of the present disclosure have been described in connection with various embodiments, it will be understood that the apparatus and methods of the present disclosure are capable of further modifications. This application is intended to cover any variations, uses, or adaptations of the apparatus and methods following, in general, the principles of the present disclosure, and including such departures from the present disclosure as, within the known and customary practice within the art to which the disclosure pertains.

(66) Furthermore, it is noted that the construction and arrangement of the disclosed long-term medicament storage apparatus, and their various components and assemblies, as shown in the various exemplary embodiments is illustrative only. Although only a few embodiments of the subject matter at issue have been described in detail in the present disclosure, those skilled in the art who review the present disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed may be constructed of multiple parts or elements, and vice versa. Also, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims. Furthermore, the order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present disclosure.

(67) It will be appreciated that the systems and approaches described herein may be used for the storage and transport of drugs in various states, such as but not limited to drug products which have undergone completion of mixing and/or other finishing steps, drug substances which are intended to be mixed and/or finished after shipping, components or ingredients to be used in a drug, or other drug-related states or components.

(68) The above description describes various devices, assemblies, components, subsystems and methods for use related to a drug delivery device. The devices, assemblies, components, subsystems, methods or drug delivery devices can further comprise or be used with a drug including but not limited to those drugs identified below as well as their generic and biosimilar counterparts. The term drug, as used herein, can be used interchangeably with other similar terms and can be used to refer to any type of medicament or therapeutic material including traditional and non-traditional pharmaceuticals, nutraceuticals, supplements, biologics, biologically active agents and compositions, large molecules, biosimilars, bioequivalents, therapeutic antibodies, polypeptides, proteins, small molecules and generics. Non-therapeutic injectable materials are also encompassed. The drug may be in liquid form, a lyophilized form, or in a reconstituted from

lyophilized form. The following example list of drugs should not be considered as all-inclusive or limiting.

(69) The drug will be contained in a reservoir. In some instances, the reservoir is a primary container that is either filled or pre-filled for treatment with the drug. The primary container can be a vial, a cartridge or a pre-filled syringe.

(70) In some embodiments, the reservoir of the drug delivery device may be filled with or the device can be used with colony stimulating factors, such as granulocyte colony-stimulating factor (G-CSF). Such G-CSF agents include but are not limited to Neulasta® (pegfilgrastim, pegylated filgastrim, pegylated G-CSF, pegylated hu-Met-G-CSF) and Neupogen® (filgrastim, G-CSF, hu-MetG-CSF), UDENYCA® (pegfilgrastim-cbqv), Ziextenzo® (LA-EP2006; pegfilgrastim-bmez), or FULPHILA (pegfilgrastim-bmez).

(71) In other embodiments, the drug delivery device may contain or be used with an erythropoiesis stimulating agent (ESA), which may be in liquid or lyophilized form. An ESA is any molecule that stimulates erythropoiesis. In some embodiments, an ESA is an erythropoiesis stimulating protein. As used herein, "erythropoiesis stimulating protein" means any protein that directly or indirectly causes activation of the erythropoietin receptor, for example, by binding to and causing dimerization of the receptor. Erythropoiesis stimulating proteins include erythropoietin and variants, analogs, or derivatives thereof that bind to and activate erythropoietin receptor; antibodies that bind to erythropoietin receptor and activate the receptor; or peptides that bind to and activate erythropoietin receptor. Erythropoiesis stimulating proteins include, but are not limited to, Epogen® (epoetin alfa), Aranesp® (darbepoetin alfa), Dynepo® (epoetin delta), Mircera® (methyoxy polyethylene glycol-epoetin beta), Hematide®, MRK-2578, INS-22, Retacrit® (epoetin zeta), Neorecormon® (epoetin beta), Silapo® (epoetin zeta), Binocrit® (epoetin alfa), epoetin alfa Hexal, Abseamed® (epoetin alfa), Ratioepo® (epoetin theta), Eporatio® (epoetin theta), Biopoin® (epoetin theta), epoetin alfa, epoetin beta, epoetin iota, epoetin omega, epoetin delta, epoetin zeta, epoetin theta, and epoetin delta, pegylated erythropoietin, carbamylated erythropoietin, as well as the molecules or variants or analogs thereof.

(72) Among particular illustrative proteins are the specific proteins set forth below, including fusions, fragments, analogs, variants or derivatives thereof: OPGL specific antibodies, peptibodies, related proteins, and the like (also referred to as RANKL specific antibodies, peptibodies and the like), including fully humanized and human OPGL specific antibodies, particularly fully humanized monoclonal antibodies; Myostatin binding proteins, peptibodies, related proteins, and the like, including myostatin specific peptibodies; IL-4 receptor specific antibodies, peptibodies, related proteins, and the like, particularly those that inhibit activities mediated by binding of IL-4 and/or IL-13 to the receptor; Interleukin 1-receptor 1 ("IL1-R1") specific antibodies, peptibodies, related proteins, and the like; Ang2 specific antibodies, peptibodies, related proteins, and the like; NGF specific antibodies, peptibodies, related proteins, and the like; CD22 specific antibodies, peptibodies, related proteins, and the like, particularly human CD22 specific antibodies, such as but not limited to humanized and fully human antibodies, including but not limited to humanized and fully human monoclonal antibodies, particularly including but not limited to human CD22 specific IgG antibodies, such as, a dimer of a human-mouse monoclonal hLL2 gamma-chain disulfide linked to a human-mouse monoclonal hLL2 kappa-chain, for example, the human CD22 specific fully humanized antibody in Epratuzumab, CAS registry number 501423-23-0; IGF-1 receptor specific antibodies, peptibodies, and related proteins, and the like including but not limited to anti-IGF-1R antibodies; B-7 related protein 1 specific antibodies, peptibodies, related proteins and the like ("B7RP-1" and also referring to B7H2, ICOSL, B7h, and CD275), including but not limited to B7RP-specific fully human monoclonal IgG2 antibodies, including but not limited to fully human IgG2 monoclonal antibody that binds an epitope in the first immunoglobulin-like domain of B7RP-1, including but not limited to those that inhibit the interaction of B7RP-1 with its natural receptor, ICOS, on activated T cells; IL-15 specific antibodies, peptibodies, related proteins, and the like,

such as, in particular, humanized monoclonal antibodies, including but not limited to HuMax IL-15 antibodies and related proteins, such as, for instance, 145c7; IFN gamma specific antibodies, peptibodies, related proteins and the like, including but not limited to human IFN gamma specific antibodies, and including but not limited to fully human anti-IFN gamma antibodies; TALL-1 specific antibodies, peptibodies, related proteins, and the like, and other TALL specific binding proteins; Parathyroid hormone ("PTH") specific antibodies, peptibodies, related proteins, and the like; Thrombopoietin receptor ("TPO-R") specific antibodies, peptibodies, related proteins, and the like; Hepatocyte growth factor ("HGF") specific antibodies, peptibodies, related proteins, and the like, including those that target the HGF/SF:cMet axis (HGF/SF:c-Met), such as fully human monoclonal antibodies that neutralize hepatocyte growth factor/scatter (HGF/SF); TRAIL-R2 specific antibodies, peptibodies, related proteins and the like; Activin A specific antibodies, peptibodies, proteins, and the like; TGF-beta specific antibodies, peptibodies, related proteins, and the like; Amyloid-beta protein specific antibodies, peptibodies, related proteins, and the like; c-Kit specific antibodies, peptibodies, related proteins, and the like, including but not limited to proteins that bind c-Kit and/or other stem cell factor receptors; OX40L specific antibodies, peptibodies, related proteins, and the like, including but not limited to proteins that bind OX40L and/or other ligands of the OX40 receptor; Activase® (alteplase, tPA); Aranesp® (darbepoetin alfa) Erythropoietin [30-asparagine, 32-threonine, 87-valine, 88-asparagine, 90-threonine], Darbepoetin alfa, novel erythropoiesis stimulating protein (NESP); Epogen® (epoetin alfa, or erythropoietin); GLP-1, Avonex® (interferon beta-1a); Bexxar® (tositumomab, anti-CD22 monoclonal antibody); Betaseron® (interferon-beta); Campath® (alemtuzumab, anti-CD52 monoclonal antibody); Dynepo® (epoetin delta); Velcade® (bortezomib); MLN0002 (anti-α4β7 mAb); MLN1202 (anti-CCR2 chemokine receptor mAb); Enbrel® (etanercept, TNF-receptor/Fc fusion protein, TNF blocker); Eprex® (epoetin alfa); Erbitux® (cetuximab, anti-EGFR/HER1/c-ErbB-1); Genotropin® (somatropin, Human Growth Hormone); Herceptin® (trastuzumab, anti-HER2/neu (erbB2) receptor mAb); Kanjinti™ (trastuzumab-anns) anti-HER2 monoclonal antibody, biosimilar to Herceptin®, or another product containing trastuzumab for the treatment of breast or gastric cancers; Humatrope® (somatropin, Human Growth Hormone); Humira® (adalimumab); Vectibix® (panitumumab), Xgeva® (denosumab), Prolia® (denosumab), Immunoglobulin G2 Human Monoclonal Antibody to RANK Ligand, Enbrel® (etanercept, TNF-receptor/Fc fusion protein, TNF blocker), Nplate® (romiplostim), rilotumumab, ganitumab, conatumumab, brodalumab, insulin in solution; Infergen® (interferon alfacon-1); Natrecor® (nesiritide; recombinant human B-type natriuretic peptide (hBNP); Kineret® (anakinra); Leukine® (sargamostim, rhuGM-CSF); LymphoCide® (epratuzumab, anti-CD22 mAb); Benlysta™ (lymphostat B, belimumab, anti-BlyS mAb); Metalyse® (tenecteplase, t-PA analog); Mircera® (methoxy polyethylene glycol-epoetin beta); Mylotarg® (gemtuzumab ozogamicin); Raptiva® (efalizumab); Cimzia® (certolizumab pegol, CDP 870); Soliris™ (eculizumab); pexelizumab (anti-C5 complement); Numax® (MEDI-524); Lucentis® (ranibizumab); Panorex® (17-1A, edrecolomab); Trabio® (lerdelimumab); TheraCim hR3 (nimotuzumab); Omnitarg (pertuzumab, 2C4); Osidem® (IDM-1); OvaRex® (B43.13); Nuvion® (visilizumab); cantuzumab mertansine (huC242-DM1); NeoRecormon® (epoetin beta); Neumega® (oprelvekin, human interleukin-11); Orthoclone OKT3® (muromonab-CD3, anti-CD3 monoclonal antibody); Procrit® (epoetin alfa); Remicade® (infliximab, anti-TNFα monoclonal antibody); Reopro® (abciximab, anti-GP IIb/Ilia receptor monoclonal antibody); Actemra® (anti-IL6 Receptor mAb); Avastin® (bevacizumab), HuMax-CD4 (zanolimumab); Mvasi™ (bevacizumab-awwb); Rituxan® (rituximab, anti-CD20 mAb); Tarceva® (erlotinib); Roferon-A®-(interferon alfa-2a); Simulect® (basiliximab); Prexige® (lumiracoxib); Synagis® (palivizumab); 145c7-CHO (anti-IL15 antibody, see U.S. Pat. No. 7,153,507); Tysabri® (natalizumab, anti-α4integrin mAb); Valortim® (MDX-1303, anti-*B. anthracis* protective antigen mAb); ABthrax™ Xolair® (omalizumab); ETI211 (anti-MRSA mAb); IL-1 trap (the Fc portion of human IgG1 and the extracellular domains of both IL-1 receptor components (the Type I receptor and receptor accessory protein)); VEGF trap (Ig domains of VEGFR1 fused to IgG1 Fc); Zenapax® (daclizumab); Zenapax® (daclizumab, anti-IL-2Ra mAb); Zevalin® (ibritumomab tiuxetan); Zetia® (ezetimibe); Orencia® (atacicept, TACI-Ig); anti-CD80 monoclonal antibody (galiximab); anti-CD23 mAb (lumiliximab); BR2-Fc (huBR3/huFc fusion protein, soluble BAFF antagonist); CNTO 148 (golimumab, anti-TNFα mAb); HGS-ETR1 (mapatumumab; human anti-TRAIL Receptor-1 mAb); HuMax-CD20 (ocrelizumab, anti-CD20 human mAb); HuMax-EGFR (zalutumumab); M200 (volociximab, anti-α5β1 integrin mAb); MDX-010 (ipilimumab, anti-CTLA-4 mAb and VEGFR-1 (IMC-18F1); anti-BR3 mAb; anti-*C. difficile* Toxin A and Toxin B C mAbs MDX-066 (CDA-1) and MDX-1388); anti-CD22 dsFv-PE38 conjugates (CAT-3888 and CAT-8015); anti-CD25 mAb (HuMax-TAC); anti-CD3 mAb (NI-0401); adecatumumab; anti-CD30 mAb (MDX-060); MDX-1333 (anti-IFNAR); anti-CD38 mAb (HuMax CD38); anti-CD40L mAb; anti-Cripto mAb; anti-CTGF Idiopathic Pulmonary Fibrosis Phase I Fibrogen (FG-3019); anti-CTLA4 mAb; anti-eotaxin1 mAb (CAT-213); anti-FGF8 mAb; anti-ganglioside GD2 mAb; anti-ganglioside GM2 mAb; anti-GDF-8 human mAb (MYO-029); anti-GM-CSF Receptor mAb (CAM-3001); anti-HepC mAb (HuMax HepC); anti-IFNα mAb (MEDI-545, MDX-198); anti-IGF1R mAb; anti-IGF-1R mAb (HuMax-Inflam); anti-IL12 mAb (ABT-874); anti-IL12/1L23 mAb (CNTO 1275); anti-IL13 mAb (CAT-354); anti-IL2Ra mAb (HuMax-TAC); anti-IL5 Receptor mAb; anti-integrin receptors mAb (MDX-018, CNTO 95); anti-IP10 Ulcerative Colitis mAb (MDX-1100); BMS-66513; anti-Mannose Receptor/hCGβ mAb (MDX-1307); anti-mesothelin dsFv-PE38 conjugate (CAT-5001); anti-PD1mAb (MDX-1106 (ONO-4538)); anti-PDGFRα antibody (IMC-3G3); anti-TGFβ mAb (GC-1008); anti-TRAIL Receptor-2 human mAb (HGS-ETR2); anti-TWEAK mAb; anti-VEGFR/Flt-1 mAb; and anti-ZP3 mAb (HuMax-ZP3).

(73) In some embodiments, the drug delivery device may contain or be used with a sclerostin antibody, such as but not limited to romosozumab, blosozumab, BPS 804 (Novartis), Evenity™ (romosozumab-aqqg), another product containing romosozumab for treatment of postmenopausal osteoporosis and/or fracture healing and in other embodiments, a monoclonal antibody (IgG) that binds human Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9). Such PCSK9 specific antibodies include, but are not limited to, Repatha® (evolocumab) and Praluent® (alirocumab). In other embodiments, the drug delivery device may contain or be used with rilotumumab, bixalomer, trebananib, ganitumab, conatumumab, motesanib diphosphate, brodalumab, vidupiprant or panitumumab. In some embodiments, the reservoir of the drug delivery device may be filled with or the device can be used with IMLYGIC® (talimogene laherparepvec) or another oncolytic HSV for the treatment of melanoma or other cancers including but are not limited to OncoVEXGALV/CD; OrienX010; G207, 1716; NV1020; NV12023; NV1034; and NV1042. In some embodiments, the drug delivery device may contain or be used with endogenous tissue inhibitors of metalloproteinases (TIMPs) such as but not limited to TIMP-3. In some embodiments, the drug delivery device may contain or be used with Aimovig® (erenumab-aooe), anti-human CGRP-R (calcitonin gene-related peptide type 1 receptor) or another product containing erenumab for the treatment of migraine headaches. Antagonistic antibodies for human calcitonin gene-related peptide (CGRP) receptor such as but not limited to erenumab and bispecific antibody molecules that target the CGRP receptor and other headache targets may also be delivered with a drug delivery device of the present disclosure. Additionally, bispecific T cell engager (BITE®) molecules such as but not limited to BLINCYTO® (blinatumomab) can be used in or with the drug delivery device of the present disclosure. In some embodiments, the drug delivery device may contain or be used with an APJ large molecule agonist such as but not limited to apelin or analogues thereof. In some embodiments, a therapeutically effective amount of an anti-thymic stromal lymphopoietin (TSLP) or TSLP receptor antibody is used in or with the drug delivery device of the present disclosure. In some embodiments, the drug delivery device may contain or be used with Avsola™ (infliximab-axxq), anti-TNF a monoclonal antibody, biosimilar to Remicade® (infliximab) (Janssen Biotech, Inc.) or another product containing infliximab for the treatment of

autoimmune diseases. In some embodiments, the drug delivery device may contain or be used with Kyprolis® (carfilzomib), (2S)—N—((S)-1-((S)-4-methyl-1-((R)-2-methyloxiran-2-yl)-1-oxopentan-2-ylcarbamoyl)-2-phenylethyl)-2-((S)-2-(2-morpholinoacetamido)-4-phenylbutanamido)-4-methylpentanamide, or another product containing carfilzomib for the treatment of multiple myeloma. In some embodiments, the drug delivery device may contain or be used with Otezla® (apremilast), N-[2-[(1S)-1-(3-ethoxy-4-methoxyphenyl)-2-(methylsulfonyl)ethyl]-2,3-dihydro-1,3-dioxo-1H-isoindol-4-yl]acetamide, or another product containing apremilast for the treatment of various inflammatory diseases. In some embodiments, the drug delivery device may contain or be used with Parsabiv™ (etelcalcetide HCl, KAI-4169) or another product containing etelcalcetide HCl for the treatment of secondary hyperparathyroidism (sHPT) such as in patients with chronic kidney disease (KD) on hemodialysis. In some embodiments, the drug delivery device may contain or be used with ABP 798 (rituximab), a biosimilar candidate to Rituxan®/MabThera™, or another product containing an anti-CD20 monoclonal antibody. In some embodiments, the drug delivery device may contain or be used with a VEGF antagonist such as a non-antibody VEGF antagonist and/or a VEGF-Trap such as aflibercept (Ig domain 2 from VEGFR1 and Ig domain 3 from VEGFR2, fused to Fc domain of IgG1). In some embodiments, the drug delivery device may contain or be used with ABP 959 (eculizumab), a biosimilar candidate to Soliris®, or another product containing a monoclonal antibody that specifically binds to the complement protein C5. In some embodiments, the drug delivery device may contain or be used with Rozibafusp alfa (formerly AMG 570) is a novel bispecific antibody-peptide conjugate that simultaneously blocks ICOSL and BAFF activity. In some embodiments, the drug delivery device may contain or be used with Omecamtiv mecarbil, a small molecule selective cardiac myosin activator, or myotrope, which directly targets the contractile mechanisms of the heart, or another product containing a small molecule selective cardiac myosin activator. In some embodiments, the drug delivery device may contain or be used with Sotorasib (formerly known as AMG 510), a KRAS.sup.G12C small molecule inhibitor, or another product containing a KRAS.sup.G12C small molecule inhibitor. In some embodiments, the drug delivery device may contain or be used with Tezepelumab, a human monoclonal antibody that inhibits the action of thymic stromal lymphopoietin (TSLP), or another product containing a human monoclonal antibody that inhibits the action of TSLP. In some embodiments, the drug delivery device may contain or be used with AMG 714, a human monoclonal antibody that binds to Interleukin-15 (IL-15) or another product containing a human monoclonal antibody that binds to Interleukin-15 (IL-15). In some embodiments, the drug delivery device may contain or be used with AMG 890, a small interfering RNA (siRNA) that lowers lipoprotein(a), also known as Lp(a), or another product containing a small interfering RNA (siRNA) that lowers lipoprotein(a). In some embodiments, the drug delivery device may contain or be used with ABP 654 (human IgG1 kappa antibody), a biosimilar candidate to Stelara®, or another product that contains human IgG1 kappa antibody and/or binds to the p40 subunit of human cytokines interleukin (IL)-12 and IL-23. In some embodiments, the drug delivery device may contain or be used with Amjevita™ or Amgevita™ (formerly ABP 501) (mab anti-TNF human IgG1), a biosimilar candidate to Humira®, or another product that contains human mab anti-TNF human IgG1. In some embodiments, the drug delivery device may contain or be used with AMG 160, or another product that contains a half-life extended (HLE) anti-prostate-specific membrane antigen (PSMA)×anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with AMG 119, or another product containing a delta-like ligand 3 (DLL3) CART (chimeric antigen receptor T cell) cellular therapy. In some embodiments, the drug delivery device may contain or be used with AMG 119, or another product containing a delta-like ligand 3 (DLL3) CART (chimeric antigen receptor T cell) cellular therapy. In some embodiments, the drug delivery device may contain or be used with AMG 133, or another product containing a gastric inhibitory polypeptide receptor (GIPR) antagonist and GLP-1R agonist. In some embodiments, the drug delivery device may contain or be used with AMG 171 or another product containing a Growth Differential Factor 15 (GDF15) analog. In some embodiments, the drug delivery device may contain or be used with AMG 176 or another product containing a small molecule inhibitor of myeloid cell leukemia 1 (MCL-1). In some embodiments, the drug delivery device may contain or be used with AMG 199 or another product containing a half-life extended (HLE) bispecific T cell engager construct (BITE®). In some embodiments, the drug delivery device may contain or be used with AMG 256 or another product containing an anti-PD-1×IL21 mutein and/or an IL-21 receptor agonist designed to selectively turn on the Interleukin 21 (IL-21) pathway in programmed cell death-1 (PD-1) positive cells. In some embodiments, the drug delivery device may contain or be used with AMG 330 or another product containing an anti-CD33×anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with AMG 404 or another product containing a human anti-programmed cell death-1 (PD-1) monoclonal antibody being investigated as a treatment for patients with solid tumors. In some embodiments, the drug delivery device may contain or be used with AMG 427 or another product containing a half-life extended (HLE) anti-fms-like tyrosine kinase 3 (FLT3)×anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with AMG 430 or another product containing an anti-Jagged-1 monoclonal antibody. In some embodiments, the drug delivery device may contain or be used with AMG 506 or another product containing a multi-specific FAP×4-1BB-targeting DARPin® biologic under investigation as a treatment for solid tumors. In some embodiments, the drug delivery device may contain or be used with AMG 509 or another product containing a bivalent T-cell engager and is designed using XmAb® 2+1 technology. In some embodiments, the drug delivery device may contain or be used with AMG 562 or another product containing a half-life extended (HLE) CD19×CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with Efavaleukin alfa (formerly AMG 592) or another product containing an IL-2 mutein Fc fusion protein. In some embodiments, the drug delivery device may contain or be used with AMG 596 or another product containing a CD3×epidermal growth factor receptor vIII (EGFRvIII) BiTE® (bispecific T cell engager) molecule. In some embodiments, the drug delivery device may contain or be used with AMG 673 or another product containing a half-life extended (HLE) anti-CD33×anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with AMG 701 or another product containing a half-life extended (HLE) anti-B-cell maturation antigen (BCMA)×anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with AMG 757 or another product containing a half-life extended (HLE) anti-delta-like ligand 3 (DLL3)×anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the drug delivery device may contain or be used with AMG 910 or another product containing a half-life extended (HLE) epithelial cell tight junction protein claudin 18.2×CD3 BiTE® (bispecific T cell engager) construct.

(74) Although the drug delivery devices, assemblies, components, subsystems and methods have been described in terms of exemplary embodiments, they are not limited thereto. The detailed description is to be construed as exemplary only and does not describe every possible embodiment of the present disclosure. Numerous alternative embodiments could be implemented, using either current technology or technology developed after the filing date of this patent that would still fall within the scope of the claims defining the invention(s) disclosed herein.

(75) Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above described embodiments without departing from the spirit and scope of the invention(s) disclosed herein, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept(s).

Claims

1. An apparatus for storing a medicament delivery device, the apparatus comprising: a container defining a storage cavity having an opening; a syringe pre-filled with medicament and disposed in the storage cavity; an absorbent disposed in the storage cavity, the absorbent at least partially hydrated with a liquid solution; and a seal member at least selectively connected with the storage cavity to form a seal that is at least substantially gas impermeable.
2. The apparatus of claim 1, wherein the cavity of the container and its contents are sterile.
3. The apparatus of claim 1, further comprising a gas permeable seal member disposed on the container over at least part of the opening enclosing at least part of the cavity containing the syringe, the gas permeable seal member disposed between the seal member and the container.
4. The apparatus of claim 3, wherein the cavity comprises a first sub-cavity and a second sub-cavity separated by a container wall, the first sub-cavity containing the syringe and the second sub-cavity containing the absorbent.
5. The apparatus of claim 4, wherein the gas-permeable seal member is disposed over the first sub-cavity but not the second sub-cavity, and the seal member is disposed over the first and second sub-cavities.
6. The apparatus of claim 1, wherein the absorbent is at least partially hydrated with a water solution.
7. The apparatus of claim 1, wherein the seal member is a sealed gas-impermeable bag in which the container, the syringe, and the absorbent are disposed.
8. A method of packaging a medicament delivery device for storage, the method comprising: providing a container defining a storage cavity and an opening; disposing a syringe pre-filled with a medicament in the storage cavity of the container; disposing an absorbent in the storage cavity of the container, the absorbent being at least partially hydrated with a liquid solution; and sealing the storage cavity of the container with a seal member by at least selectively connecting the seal member with cavity to form a seal that is at least substantially gas impermeable.
9. The method of claim 8, wherein sealing the storage cavity of the container comprises sealing the seal member to the container adjacent to the opening to enclose the storage cavity.
10. The method of claim 8, further comprising sterilizing the storage cavity and the syringe prior to sealing the storage cavity of the container with the seal member.
11. The method of claim 10, wherein sterilizing the storage cavity and the syringe comprises exposing the storage cavity and the syringe to a vapor phase hydrogen peroxide (VPHP) sterilizing process.
12. The method of claim 10, further comprising sealing at least part of the storage cavity containing the syringe with a gas-permeable seal member prior to sealing the storage cavity of the container with the seal member.
13. The method of claim 8, wherein sealing the storage cavity of the container comprises sealing the container in a sealed gas-impermeable bag.
14. The method of claim 8, wherein disposing the syringe in the storage cavity of the container comprises disposing the syringe in a first sub-cavity of the container, and disposing the absorbent in the storage cavity comprises disposing the absorbent in a second sub-cavity of the container, the first and second sub-cavities separated by a container wall.
15. The method of claim 8, further comprising hydrating the absorbent in a water solution for a predetermined period of time prior to disposing the absorbent in the cavity.